

Hapticus: A Haptic Customization Tool for Effective Physical Skill Learning

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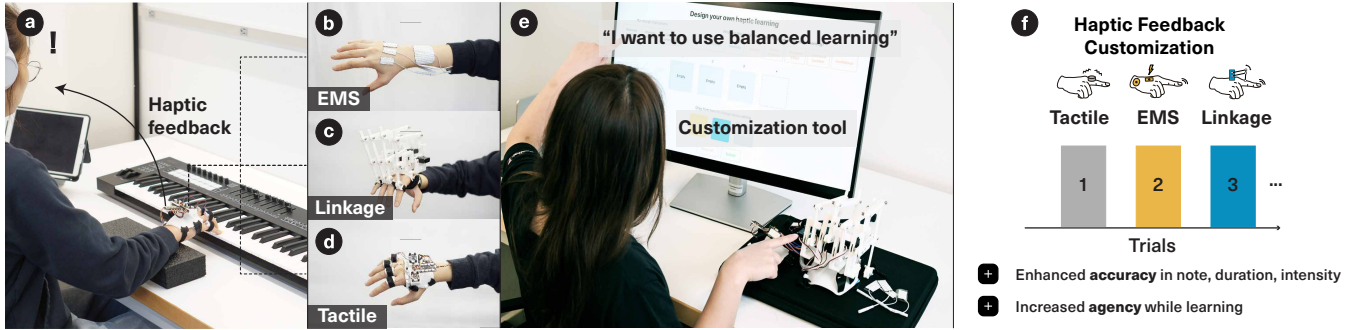


Figure 1: (a) We explored motor learning experience with multiple haptic feedback during piano learning. The combined device includes (b) a linkage-based exoskeleton, (c) a vibrotactile glove, and (d) an EMS. (e–f) We then provided a haptic customization tool, called Hapticus, with presets synthesized from performance and subjective datasets, enabling users to identify the optimal haptic feedback for their motor learning. Through this customization, users can improve their skills more effectively by leveraging the benefits of each feedback modality.

Abstract

Numerous haptic devices have been proposed to support motor learning, and haptic feedback has been shown to be effective in improving learning performance. Furthermore, recent studies have found that customized combinations of multiple haptic modalities can enhance learning outcomes and the overall experience compared to using a single modality. To enable effective haptic customization, we developed a customization tool, called Hapticus, and a combined device that integrates three modalities: Electrical Muscle Stimulation (EMS), vibrotactile feedback (Tactile), and a mechanical linkage-based exoskeleton (Linkage). These tools allow users to experience greater agency and improved feedback comfort during motor skill learning.

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1 Introduction

A wide range of motor skills is integral to everyday life, from basic tasks such as walking to complex tasks such as surgical procedures and playing musical instruments. However, developing dexterous motor skills is challenging to achieve independently as motor skill learning requires mastering multidimensional aspects, including the precise application of force, dexterity, timing, and directional control. For instance, not only does a piano learner need to understand the musical notes and sequences, but also physically coordinate their fingers to apply the correct amount of force at the right times, on the right keys.

Although other forms of fine-motor skill improvement devices have been used, such as visually providing related body movements by using flat screens, projectors, or head-mounted displays from one user to another. While visual feedback is excellent at demonstrating where and how body parts should move, it cannot directly convey essential elements of motor skills, such as muscle force or coordination.

To address this, haptic feedback systems have been proposed. Examples include the use of vibrotactile motors, exoskeletons with mechanical linkages, and electrical muscle stimulation (EMS) that can physically actuate muscles through electrodes attached to the skin. Haptic feedback that directly affects the physical actions of a user has proven to be effective in scenarios that require the learning of fine motor movement, such as playing musical instruments, sports skills, or rehabilitation. These systems demonstrated a number of positive effects on motor learning efficacy, such as improved skill performance [Nishida et al. 2022] and a more engaged experience [Turakhia et al. 2021].

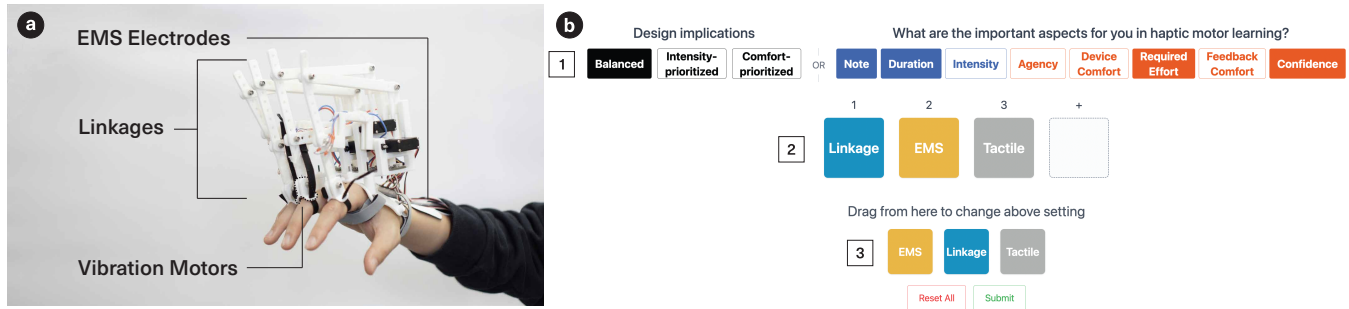


Figure 2: (a) Combined haptic device that provides three different haptic modalities. (b) Feedback order customization form

However, it remains unclear how these haptic feedback modalities differ in terms of performance and subjective measures, and it is a challenging task to customize these haptic feedback due to the complexity of learning and individual differences in preference. To address this, we propose a haptic customization tool, called *Hapticus*, that offers several presets, enabling users to easily personalize and adjust the sequence of feedback, along with a combined haptic device that integrates three modalities: EMS, vibrotactile feedback (Tactile), and a mechanical linkage-based exoskeleton (Linkage).

2 Hapticus Implementation

We developed a device that provides the three haptic modalities, all of which are commonly used in motor learning research. To enable seamless transitions between multiple haptic feedback types, we designed an integrated device as shown in Figure 2 (a). The device weighs 309 g and is controlled by Arduino Nano 33 BLE.

Exoskeleton. The device features four four-bar linkages, servo motors (FEETECH, FT1117M), and finger sockets mounted on a palm base that were 3D-printed using a PLA filament. The servo motors are powered by a 6V power supply unit. The actuated angle for the exoskeleton is set at 40 degrees. For intensity control, the speed of the motors is adjusted. The average actuation latency from stimulus to playing note is 108 ms (SD = 19.2 ms).

EMS. The EMS device consists of five electrodes. The electrodes, each measuring 1 x 3 cm, are placed on the user's back of hand with negative electrodes and wrist with a positive electrode. The device provides an adjustable pulse width from 0 to 300 μ s and operates at a stimulation frequency of 100Hz. It consists of gate-switching devices and a DC-DC converter generating an output amplitude of 35V from AAA batteries. We mainly referenced a previous study on EMS finger actuation [Takahashi et al. 2021], including aspects such as electrode placement and size. The average actuation latency from stimulus to playing note is 32 ms (SD = 8.3 ms).

Vibrotactile glove. We prepared a vibrotactile glove with four linear resonant actuators (LRAs; Vybronic, VG0832022D). Each motor operates at an approximate vibration frequency of 235Hz. These motors are securely placed in 3D-printed sockets and attached using velcro on top of each user's finger for easy adjustment and stability. They are controlled by four motor drivers (Texas Instruments, DRV2605) and powered by a pair of AAA batteries (3V).

Haptic Customization Tool. We developed a haptic customization tool (Figure 2b) based on the results of our user study [Lee et al. 2025]. The tool allows users to select their preferred design implication (Figure 2b-1). They can choose from three predefined learning styles: **balanced**, **intensity-prioritized**, or **comfort-prioritized learning**. The highlighted factors on the right indicate the expected benefits when using the selected style. Based on this input, the system recommends the most suitable haptic customization style (Figure 2b-2). Users can also increase the number of learning trials or make additional customizations (Figure 2b-3).

3 Visitor Experience

Our demo scenario features **tailored piano skill learning with multi-haptic feedback**. First, the customization tool is provided via a tablet, allowing visitors to select one of the presets or design their own learning style. Next, they wear the combined haptic device. Finally, they receive customized haptic feedback and experience personalized haptic learning with the benefits of enhanced accuracy in note, duration, intensity, and agency. The default number of trials is set to three, with the option to add additional trials.

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